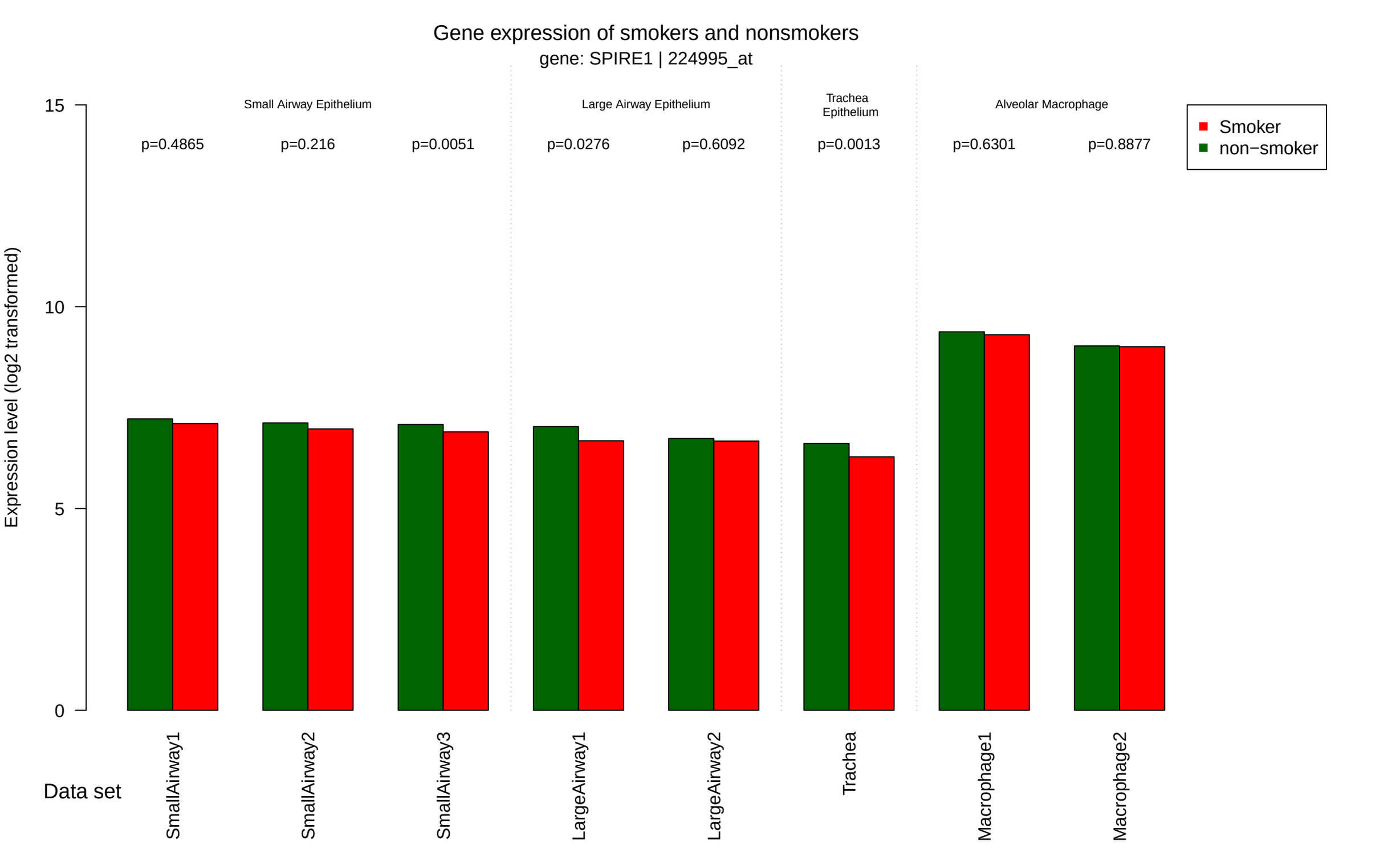
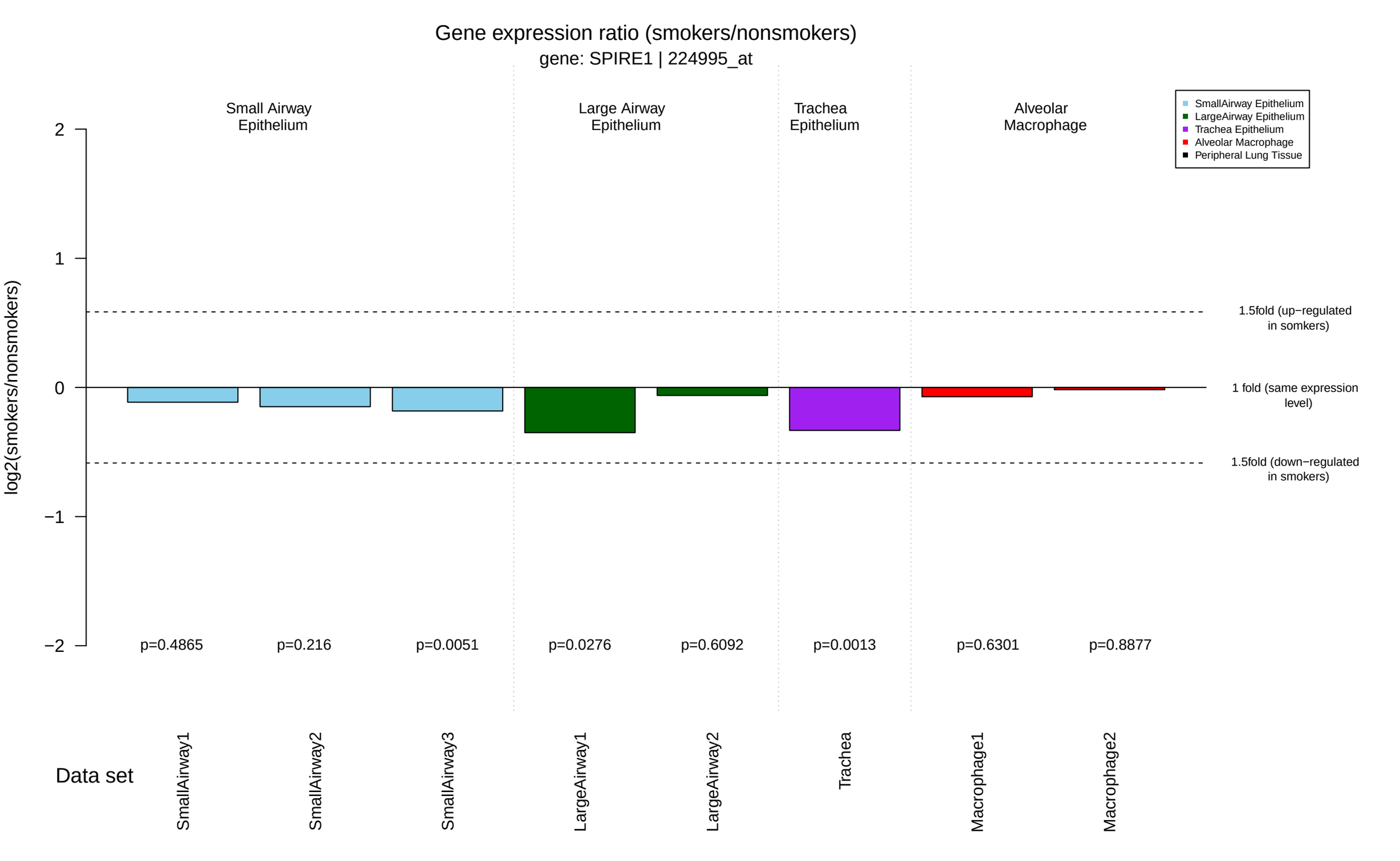
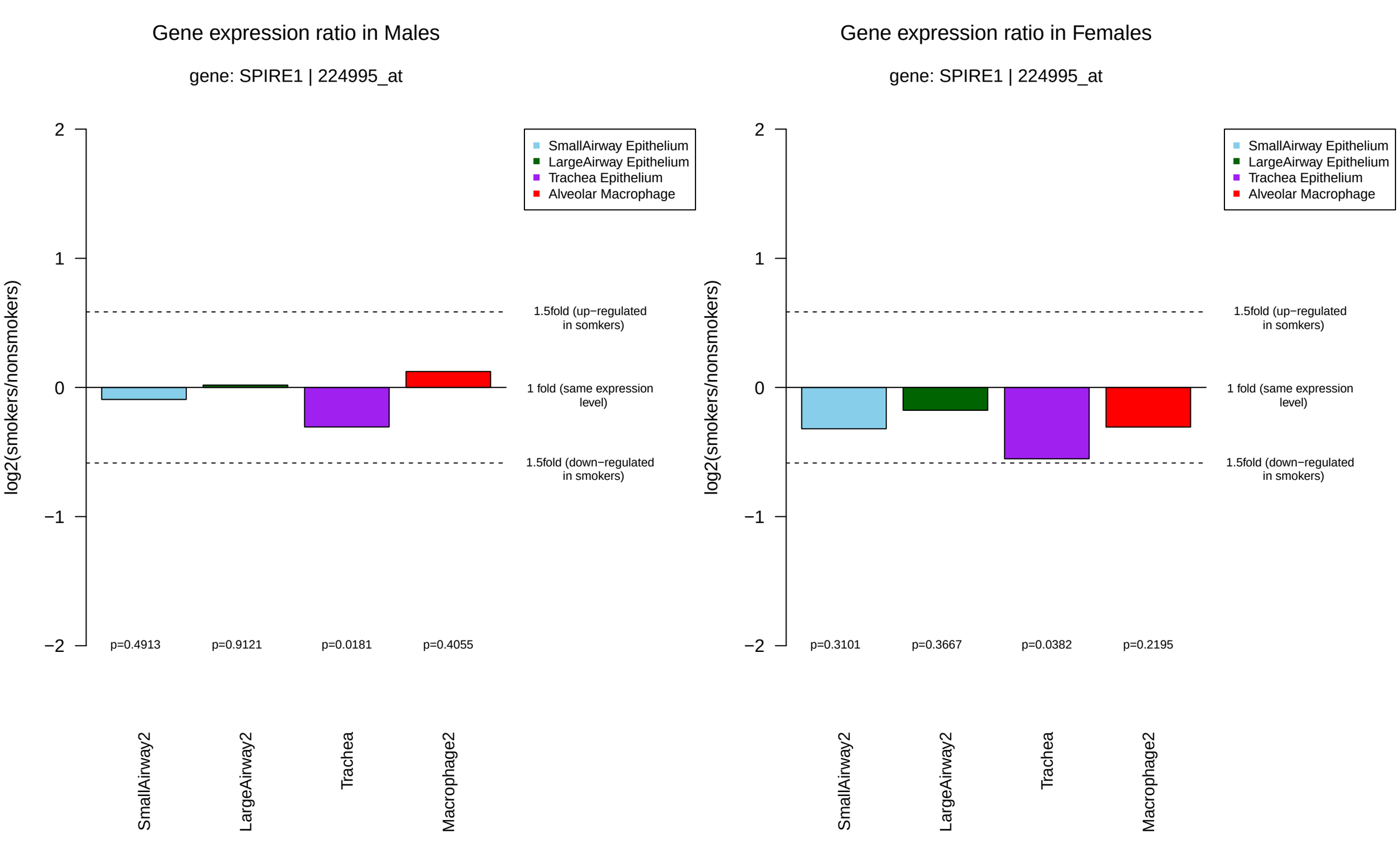
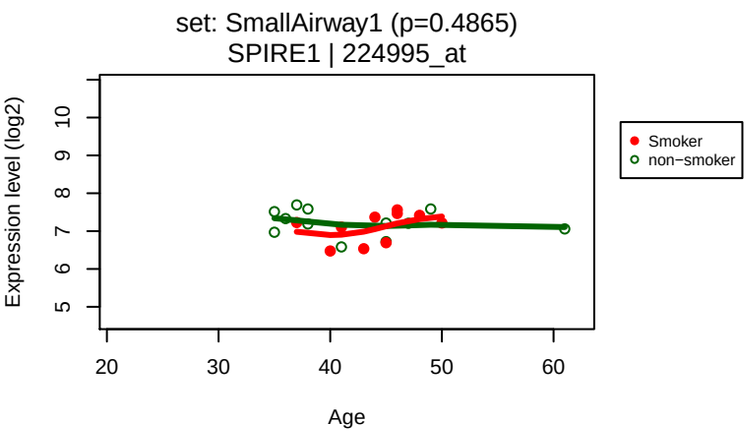
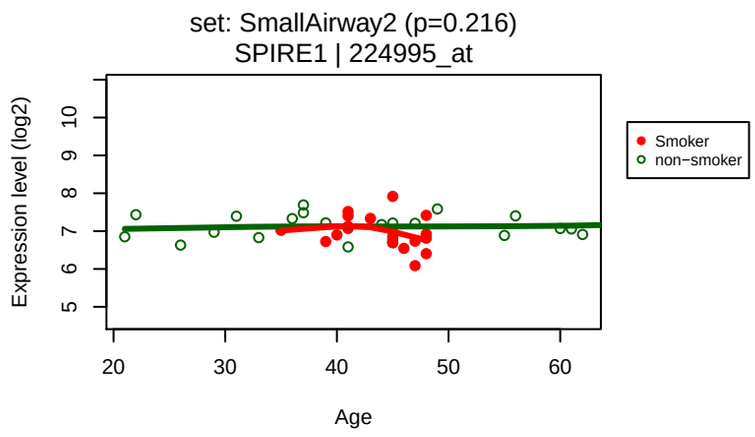
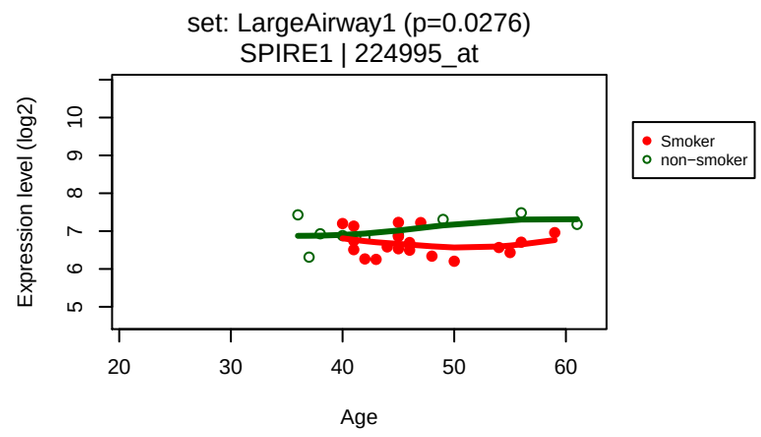

Gene expression of smokers and nonsmokers
gene: SPIRE1 | 224995_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4865
p=0.216
p=0.0051
p=0.0276
p=0.6092
p=0.0013
p=0.6301
p=0.8877
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

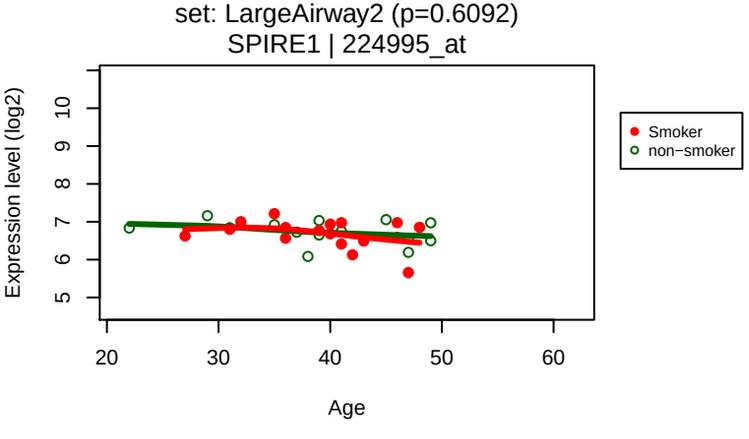

Gene expression ratio (smokers/nonsmokers)
gene: SPIRE1 | 224995_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4865
p=0.216
p=0.0051
p=0.0276
p=0.6092
p=0.0013
p=0.6301
p=0.8877
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

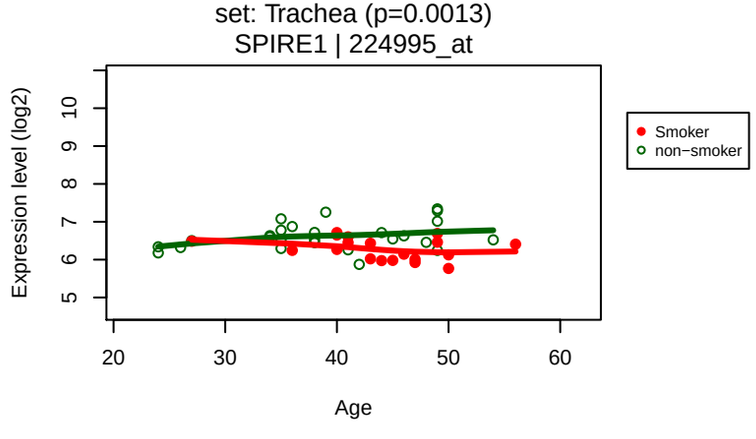

Gene expression ratio in Males
gene: SPIRE1 | 224995_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4913
p=0.9121
p=0.0181
p=0.4055
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPIRE1 | 224995_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3101
p=0.3667
p=0.0382
p=0.2195
SmallAirway2
LargeAirway2
Trachea
Macrophage2

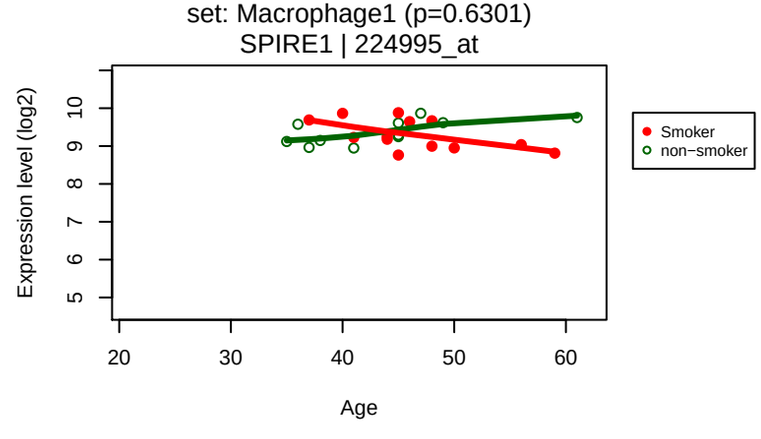

set: SmallAirway1 (p=0.4865)
SPIRE1 | 224995_at
Expression level (log2)
Age
Smoker
non−smoker

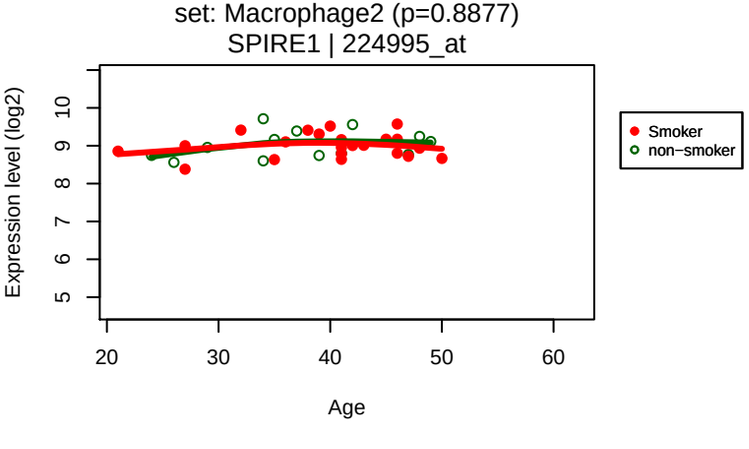

set: SmallAirway2 (p=0.216)
SPIRE1 | 224995_at
Expression level (log2)
Age
Smoker
non−smoker

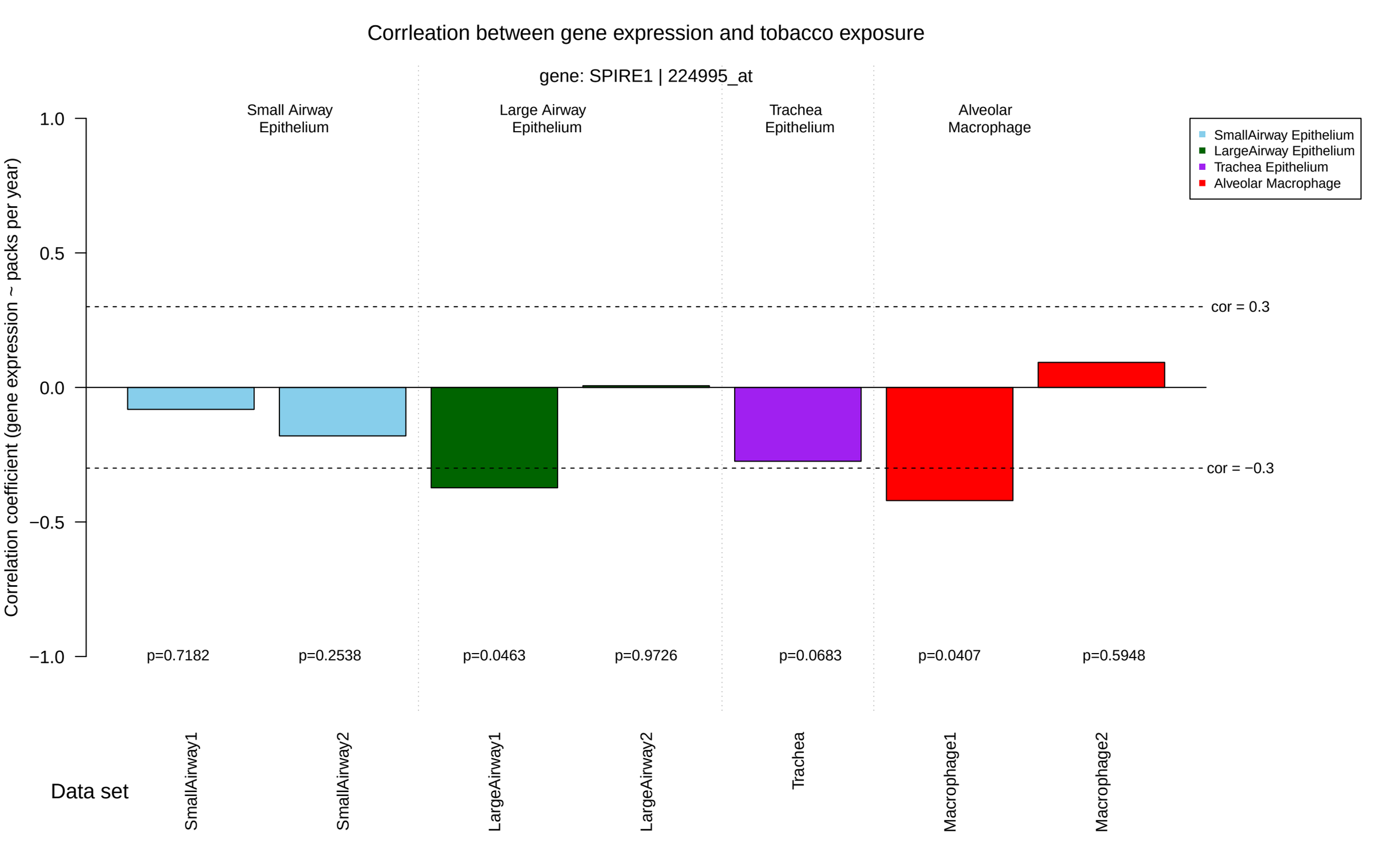

set: LargeAirway1 (p=0.0276)
SPIRE1 | 224995_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6092)
SPIRE1 | 224995_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
SPIRE1 | 224995_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6301)
SPIRE1 | 224995_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8877)
SPIRE1 | 224995_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPIRE1 | 224995_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7182
p=0.2538
p=0.0463
p=0.9726
p=0.0683
p=0.0407
p=0.5948
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2